

Clinical herd health, farm management and antimicrobial resistance in Campylobacter coli on finishing pig farms in Switzerland.

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The world-wide increase of antimicrobial resistance in micro-organisms complicates medical treatment of infected humans. We did a risk-factor analysis for the prevalence of antimicrobial resistant *Campylobacter coli* on 64 Swiss pig finishing farms. Between May and November 2001, 20 faecal samples per farm were collected from the floor of pens holding finishing pigs shortly before slaughter. Samples were pooled and cultured for *Campylobacter* species. Isolated *Campylobacter* strains were tested for resistance against selected antimicrobials. Additionally, information on herd health and management aspects was available from another study. Because data quality on the history of antimicrobial use on the farms was poor, only non-antimicrobial risk factors could be analysed. Statistical analyses were performed for resistance against ciprofloxacin, erythromycin, streptomycin, tetracycline, and for multiple resistance, which was defined as resistance to three or more antimicrobials. Risk factors for these outcomes--corrected for dependency of samples at herd level--were analysed in five generalised estimation-equation models. Prevalence of antimicrobial resistance among *Campylobacter* isolates was ciprofloxacin 26.1%, erythromycin 19.2%, streptomycin 78.0%, tetracycline 9.4%, and multiple resistance 6.5%. Important risk factors contributing to the prevalence of resistant strains were shortened tails, lameness, skin lesions, feed without whey, and ad libitum feeding. Multiple resistance was more likely in farms which only partially used an all-in-all-out system (OR = 37), or a continuous-flow system (OR = 3) compared to a strict all-in-all-out animal-flow. Presence of lameness (OR = 25), ill-thrift (OR = 15), and scratches at the shoulder (OR = 5) in the herd also increased the odds for multiple resistance. This study showed that on finishing farms which maintained a good herd health status and optimal farm management, the prevalence of antimicrobial resistance was also more favourable.